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Sensor-Based Infusion Monitoring and State Estimation using a Kalman Filter

Sarah Sophie Kruschinski
Eggerstedtstraße 78
22765 Hamburg
Sarah.S.Kruschinski@stud.leuphana.de

Matrikeln.: 4004023
B.Sc. Wirtschaftsinformatik
B.A. Digital Media
Minor: BWL

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A Combined Hardware–Software Project

Author: Sarah Kruschinski

Program: Digital Media & Business Informatics (Leuphana University)

Course Context:

- TechBasics II (Arduino Project)
- Fundamentals of Statistics with Introduction to Kalman Filter (MATLAB)

This report presents a structured and reflective analysis of the project.

The complete development process including source code, intermediate iterations, and visual documentation (images and video recordings), is available in the accompanying GitHub repository:

<https://github.com/sasokru/FinalProject25-26.git>

While the repository captures the experimental and iterative nature of the work, this document focuses on extracting the underlying system logic, design decisions, and conceptual insights.

1. Introduction

Infusion therapy is a fundamental component of clinical care, yet its monitoring often relies on manual observation and estimation. In high-demand environments, such as hospital wards, this introduces variability, delayed reactions, and potential risk.

This project explores how a sensor-based system combined with probabilistic state estimation can improve the reliability of infusion monitoring. Specifically, it investigates how noisy, discrete measurements—in this case, infrared-based drop detection—can be transformed into a continuous and stable estimate of flow rate.

The work is situated at the intersection of:

- embedded systems (Arduino-based sensing)
- signal processing under uncertainty
- and digital health applications

A central challenge emerged early in development:

the physical measurement process proved inherently unreliable. Optical interference within the drip chamber led to persistent false detections, revealing a fundamental gap between measured signals and the underlying physical reality. Rather than treating this as a purely technical malfunction, the project reframes it as a modeling problem:

How can we estimate the true state of a system when measurements are incomplete and corrupted by noise?

To address this, a Kalman filter was introduced as a mathematical framework for recursive state estimation. Although originally developed in a separate course context, its integration into this project proved conceptually essential rather than optional.

This report therefore presents not only the development of a functional prototype, but a shift in perspective:

from building a system that measures to building a system that interprets uncertain reality.

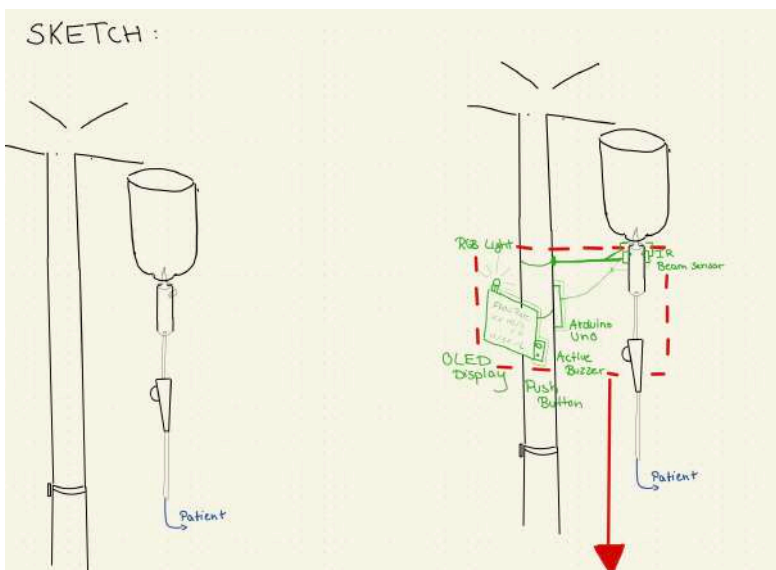


Figure 1: The first sketches.



Figure 2: The final prototype.

2. Project Context and Conceptual Framing

This project sits at the intersection of:

- Embedded systems (Arduino)
- Signal processing
- Healthcare technology
- Mathematical modeling

Two parallel tracks were developed:

2.1 Hardware System (TechBasics II)

- Infrared (IR) break beam sensor detects drops
- Arduino processes signals
- OLED display outputs flow rate
- RGB LED indicates system state

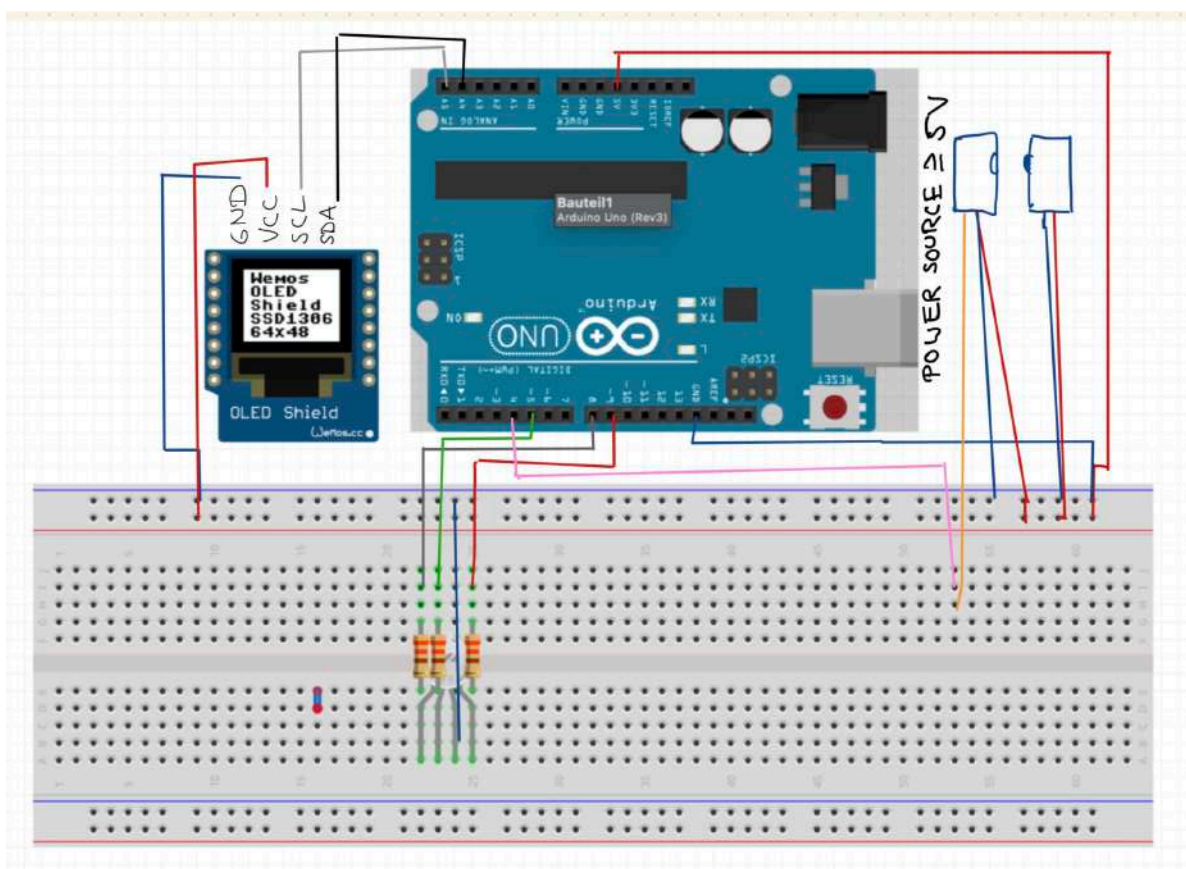


Figure 3: The electronic circuit.

2.2 Mathematical Model (Kalman Filter Project)

- MATLAB-based signal smoothing
- State estimation of flow rate
- Noise reduction and stabilization

The system consists of:

- Infusion stand with NaCl solution
- Standard infusion set with drip chamber
- IR break beam sensor aligned across the chamber
- Arduino Uno microcontroller
- OLED display (I2C, address 0x3C)
- RGB LED for status feedback

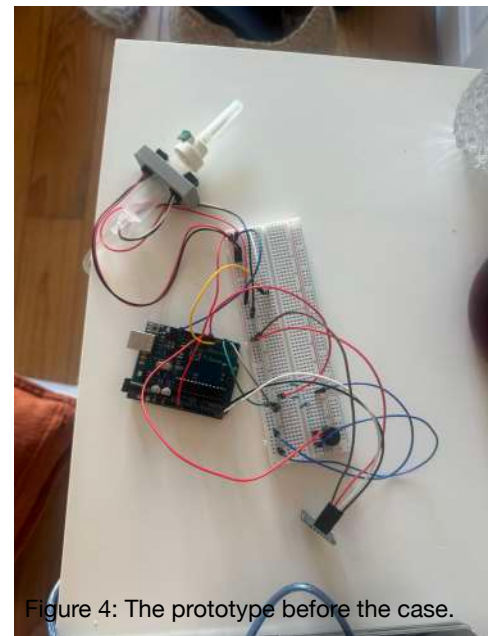


Figure 4: The prototype before the case.

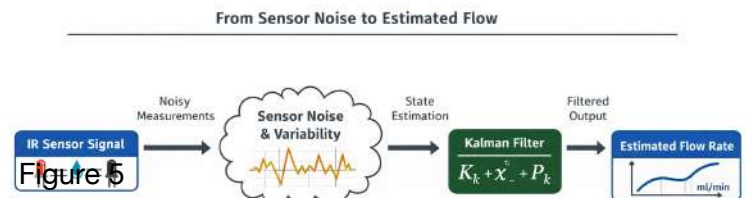
3. Measurement Principle

The measurement approach is based on optical interruption using an infrared (IR) break beam sensor positioned across the drip chamber of a standard infusion set.

As individual drops pass through the chamber, they briefly interrupt the IR beam, resulting in discrete changes in the digital output signal of the sensor. These signal transitions are interpreted as individual drop events and counted over time.

To derive a volumetric flow rate from these discrete measurements, a commonly used clinical approximation is applied, in which 20 drops correspond to approximately 1 milliliter of fluid. On this basis, the system converts drop counts into continuous flow-related metrics, such as drops per minute and milliliters per minute.

While this approach offers a simple and cost-effective method for flow estimation, its reliability depends heavily on the stability and interpretability of the underlying sensor



signal. As the following sections demonstrate, this assumption does not fully hold under real-world conditions.

3.1 Flow Rate Calculation Model

The estimation of the infusion flow rate is based on discrete drop detection over time. Let $N(t)$ denote the number of detected drops within a given time interval t .

The drop rate can then be expressed as:

$$r = N(t) / t$$

Using the calibration factor of approximately 20 drops per milliliter, the volumetric flow rate Q can be derived as:

$$Q = N(t) / (20 \cdot t)$$

This formulation provides a direct mapping from discrete event counts to a continuous flow estimate. However, the accuracy of this model depends on the reliability of drop detection, which is affected by measurement noise and sensor instability.

4. Implementation Process and Iterative Development

The development of the system followed an iterative and experiment-driven process, in which initial assumptions were continuously evaluated and refined based on observed behavior.

4.1 Initial Assumptions

At the outset, it was assumed that the IR break beam sensor would reliably detect individual drops through the transparent plastic chamber of the infusion set. Under ideal



Figure 6: The OLED display output.

conditions, each drop was expected to produce a distinct and temporally isolated interruption of the beam.

4.2 Observed Behavior

In practice, the sensor output deviated significantly from this expectation. Instead of producing discrete signals corresponding to individual drop events, the system frequently generated a continuous high-state signal. This resulted in persistent false detections and unrealistically high drop counts, indicating a fundamental mismatch between the measurement principle and the physical setup.

4.3 Source of Error

The observed behavior can be attributed primarily to optical interference within the drip chamber. The curved plastic surface, combined with the presence of liquid, introduces refraction and scattering effects that distort the infrared beam. As a consequence, the sensor is unable to clearly distinguish between interrupted and uninterrupted states, leading to unstable and ambiguous signal outputs.

4.4 Iterative Adjustments

In response to these issues, multiple adjustments were explored throughout the development process. The sensor was repeatedly repositioned and realigned in an attempt to achieve a more stable signal. In parallel, modifications were made to the signal interpretation logic within the Arduino code, particularly with respect to threshold behavior and event detection. Physical changes to the setup were also considered in order to reduce optical distortion within the measurement area. Throughout this process, the system output was continuously monitored via the serial interface, enabling real-time debugging and evaluation of incremental changes.

Despite these efforts, the signal remained inherently noisy and unreliable. This phase of the project led to a critical shift in perspective: the primary challenge was no longer the detection of drops, but the interpretation of an unstable measurement signal.

5. Signal Characteristics and Noise Analysis

The raw sensor signal exhibited persistent instability, which significantly limited its direct usability. In many cases, the output remained in a constant high state, independent of actual drop events. In addition, irregular fluctuations were observed that did not correspond to any physical changes in the system. These effects were highly sensitive to environmental conditions and material properties, particularly the optical characteristics of the drip chamber.

As a result, a clear discrepancy emerged between the measured signal and the actual infusion process. From a systems perspective, this can be understood as a classical measurement problem in which the observable signal is corrupted by noise and does not directly represent the true system state.

This realization motivated a transition from a purely hardware-based approach to a model-based interpretation of the system.

6. Kalman Filter Integration

6.1 Motivation

Given the limitations of the raw sensor data, a method was required to extract meaningful information from a noisy and unreliable measurement process. The Kalman filter provides a suitable framework for this task, as it enables the recursive estimation of a system's internal state by combining prior knowledge with noisy observations.

6.2 Conceptual Role in the System

Within this project, the Kalman filter does not replace the sensor, but rather serves as an interpretative layer. The system can therefore be understood as consisting of two interacting components: a measurement layer, in which the IR sensor produces discrete and noisy signals, and an estimation layer, in which these signals are processed to reconstruct a stable representation of the underlying flow rate.

This separation reflects a fundamental shift from direct measurement to probabilistic estimation.

6.3 System Interpretation

In this framework, the true infusion flow rate represents a hidden system state that cannot be measured directly. The sensor signal provides only an indirect and noisy observation of this state. The Kalman filter combines these elements through a recursive prediction–correction process, producing a continuously updated estimate that is significantly more stable and reliable than the raw measurements.

6.4 Mathematical Formulation of the Kalman Filter

The Kalman filter models the system using a state-space representation.

State equation:

$$x_k = A x_{k-1} + B u_k + w_k$$

Measurement equation:

$$z_k = C x_k + v_k$$

Prediction step:

$$\hat{x}_{k^-} = A \hat{x}_{k-1} + B u_k$$

$$P_{k^-} = A P_{k-1} A^T + Q$$

Update step:

$$K_k = P_{k^-} C^T (C P_{k^-} C^T + R)^{-1}$$

$$\hat{x}_k = \hat{x}_{k^-} + K_k (z_k - C \hat{x}_{k^-})$$

$$P_k = (I - K_k C) P_{k^-}$$

6.5 Discrete-Time Estimation Context

The system can be interpreted as a discrete-time stochastic process, where measurements are obtained at fixed time intervals. The Kalman filter operates recursively, updating the state estimate at each time step based on new observations.

Measurement residual:

$$y_k = z_k - C \hat{x}_k^-$$

This term represents the difference between the predicted measurement and the actual observed value. It is used to correct the state estimate and plays a central role in the filtering process.

In this project, the system dynamics are assumed to be approximately constant over short time intervals, allowing the state transition matrix A to be modeled as an identity matrix.

7. MATLAB Implementation

The Kalman filter was implemented in MATLAB using a state-space representation of the system. This required defining the system dynamics through appropriate matrices and verifying key properties such as observability. In addition, observer gains were computed to enable state estimation.

The implementation process revealed several practical challenges. These included dependencies on specific MATLAB toolboxes, as well as the difficulty of aligning the abstract mathematical model with the behavior of the physical system. Parameter tuning, particularly with respect to noise covariance matrices, proved to be a critical step in achieving meaningful and stable results.

Despite these challenges, the implementation successfully demonstrated how recursive filtering can improve the quality of noisy measurement data.

8. Results

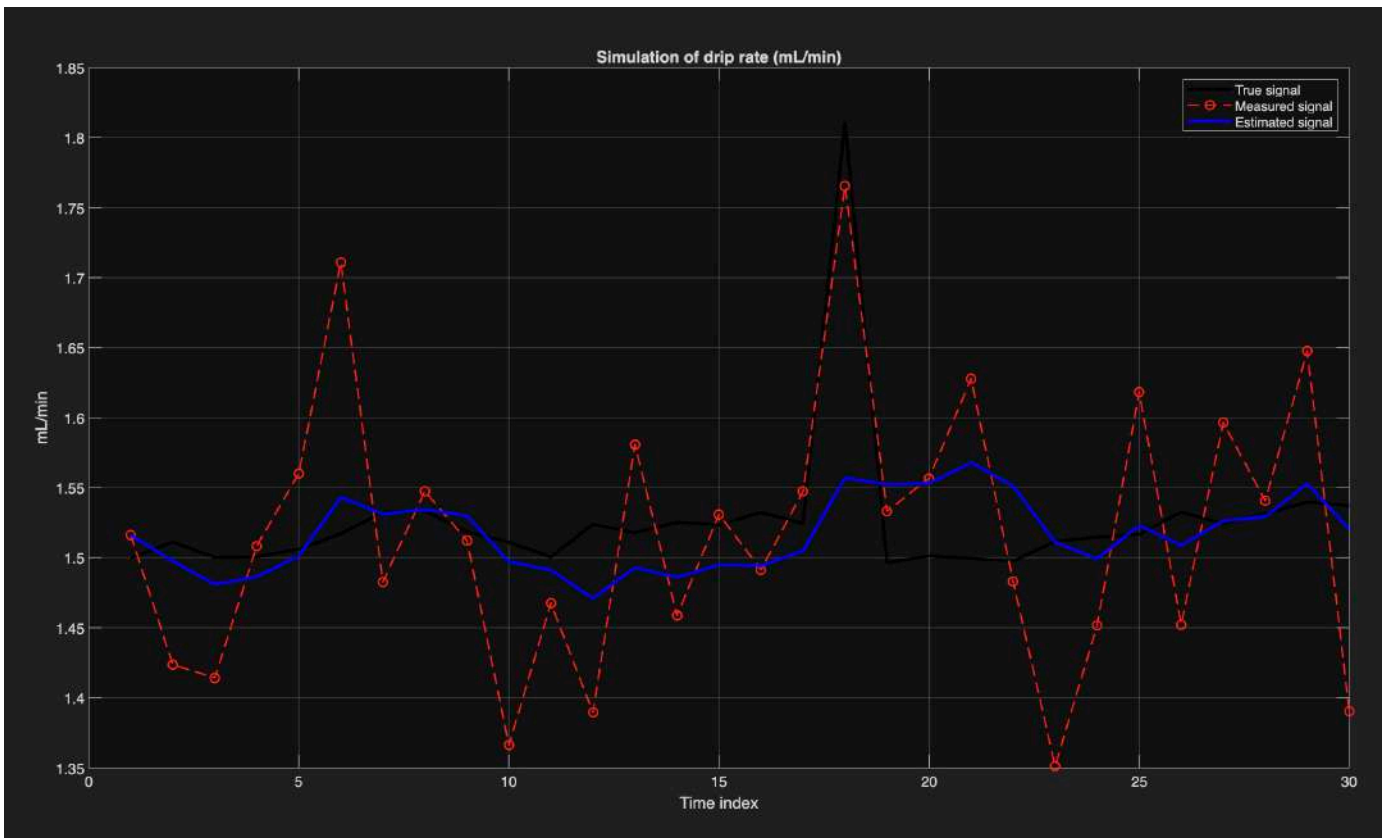


Figure 7

The Arduino-based system successfully captured signal changes from the IR sensor and enabled real-time counting of drop events. The measured values were displayed via an OLED interface, and system states were communicated through an RGB LED. However, due to the instability of the sensor signal, the raw measurements alone did not provide a reliable representation of the actual infusion flow.

The application of the Kalman filter significantly improved the interpretability of the data. Signal fluctuations were reduced, and unrealistic spikes were smoothed, resulting in a more stable and continuous estimate of the flow rate. This demonstrates that model-based estimation can effectively compensate for limitations in sensor reliability.

8.1 Simulation-Based Results

To evaluate the effect of Kalman filtering on noisy drip-rate measurements, a simulation was carried out in MATLAB. The resulting plot shows three signals: the true underlying signal, the noisy measured signal, and the estimated signal obtained through Kalman filtering.

In contrast to the earlier conceptual flow-rate example, the present section refers specifically to the MATLAB simulation shown in Figure X. The values below correspond approximately to the plotted curves and are used to support the interpretation of the simulation results.

The simulated drip rate is expressed in milliliters per minute (ml/min). The true signal remains relatively stable over time, while the measured signal exhibits visible noise and short-term deviations. The estimated signal follows the general course of the true signal while reducing the effect of these fluctuations.

| Time Index k | True Signal (ml/min) | Measured Signal z_k (ml/min) | Estimated Signal $\hat{x}_k$ (ml/min) |

1	1.50	1.52	1.51	
2	1.51	1.42	1.49	
3	1.50	1.41	1.48	
4	1.50	1.51	1.49	
5	1.51	1.56	1.50	
6	1.52	1.71	1.54	
7	1.53	1.48	1.53	
8	1.53	1.55	1.53	
9	1.52	1.51	1.53	
10	1.51	1.37	1.50	
11	1.50	1.47	1.49	
12	1.52	1.39	1.47	
13	1.52	1.58	1.49	
14	1.53	1.46	1.49	

15	1.53	1.53	1.49	
16	1.54	1.49	1.49	
17	1.53	1.55	1.50	
18	1.81	1.77	1.56	
19	1.50	1.53	1.55	
20	1.50	1.56	1.55	
21	1.50	1.63	1.57	
22	1.50	1.48	1.55	
23	1.51	1.35	1.51	
24	1.51	1.45	1.50	
25	1.52	1.62	1.52	
26	1.53	1.45	1.51	
27	1.52	1.60	1.53	
28	1.53	1.54	1.53	
29	1.54	1.65	1.55	
30	1.54	1.39	1.52	

8.2 General Characteristics of the Simulation

The simulation shows a clear distinction between the three signals. The true signal remains comparatively stable and represents the underlying drip-rate behavior of the system. The measured signal, by contrast, is affected by visible fluctuations that can be interpreted as measurement noise. These deviations include both positive spikes and negative drops, which reflect the uncertainty introduced by the sensing process.

The estimated signal produced by the Kalman filter is noticeably smoother than the measured signal. Although it still responds to changes in the system, it does so in a moderated way. This indicates that the filter does not simply replicate the raw observations, but integrates them with the model-based prediction in order to obtain a more robust state estimate.

8.3 Comparison of True, Measured, and Estimated Signals

A closer comparison of the three curves reveals the central effect of the Kalman filter. The measured signal often deviates substantially from the true signal. This can be observed, for example, around time index 6, where the measurement rises to approximately 1.71 ml/min while the true signal remains near 1.52 ml/min. A similar effect appears around time index 18, where both the true and measured signal show a peak, but the estimated signal follows this change in a more controlled and smoothed manner. At time index 23, the measured signal falls to approximately 1.35 ml/min, whereas the estimated signal remains near 1.51 ml/min and therefore avoids overreacting to a sudden local disturbance.

These examples illustrate that the Kalman filter reduces sensitivity to isolated outliers and short-term fluctuations. Instead of directly adopting every measurement value, it balances the current observation against the prior state estimate. In this way, the resulting signal remains continuous and interpretable even in the presence of noise.

8.4 Interpretation of the Filtering Effect

From a systems perspective, the simulation demonstrates that the measured signal alone is not sufficient to reliably represent the underlying physical process. The raw observations are too strongly affected by noise to be interpreted directly as the actual drip rate. This is precisely where the Kalman filter becomes relevant.

The filter improves the result by recursively combining two sources of information: first, the prediction generated by the system model, and second, the current measurement. If a measured value differs strongly from the predicted state, the filter incorporates this discrepancy only to the extent justified by the assumed noise structure. As a result, the estimated signal remains more stable than the measured one while still tracking the overall system dynamics.

In the context of infusion monitoring, this behavior is particularly important. A monitoring system should not react excessively to every local fluctuation, especially if such fluctuations are caused by sensor noise rather than actual physiological or mechanical changes. The Kalman-filtered estimate therefore provides a more meaningful basis for system interpretation than the raw sensor signal alone.

8.5 Relation to the Project Context

The simulation results are directly relevant to the broader objective of the project. In the hardware setup, the IR-based drop detection was affected by optical interference and

unstable signal behavior. This made it difficult to derive a reliable flow estimate from the sensor output alone. The MATLAB simulation addresses this issue conceptually by showing how a noisy signal can be transformed into a more stable estimate through recursive state estimation.

Even though the Arduino-based prototype and the Kalman-filter implementation originated in different course contexts, the simulation demonstrates why their combination is conceptually coherent. The hardware system provides the measurement layer, while the Kalman filter contributes the estimation layer. Together, they form a more robust interpretation framework for drip-rate monitoring under uncertain measurement conditions.

8.6 Summary of Results

Overall, the simulation confirms that the Kalman filter significantly improves the interpretability of noisy drip-rate data. While the measured signal fluctuates strongly and contains short-term deviations, the estimated signal remains smoother and follows the underlying trend of the true signal more closely. This supports the central assumption of the project: meaningful system output is not achieved through direct measurement alone, but through the combination of sensing and model-based estimation.

9. Discussion

The results highlight a fundamental principle in system design: reliable outputs are not solely determined by sensor quality, but by how measurements are interpreted.

In this project, the hardware alone was insufficient to produce meaningful results. Only through the integration of a mathematical model could the system achieve a level of stability suitable for interpretation. The Kalman filter thus serves not merely as a technical enhancement, but as a conceptual bridge between noisy physical measurements and meaningful system behavior.

10. Limitations

Despite its successful implementation, the system remains subject to several limitations. The sensor is highly susceptible to optical interference, and the Kalman filter is not yet implemented directly on the embedded system. In addition, the calibration model is simplified and does not account for variability in drop size. Finally, the system has not been validated under controlled clinical conditions.

11. Future Work

Future work could focus on improving both the sensing mechanism and the estimation process. This includes redesigning the optical setup to reduce distortion, implementing real-time filtering directly on the microcontroller, and introducing adaptive calibration methods. In a broader context, the system could be integrated into a more comprehensive digital health monitoring framework.

12. Personal Reflection

This project was shaped by the combination of technical development and practical experience in a clinical environment. What began as the construction of a measurement device evolved into a deeper exploration of how systems represent and interpret reality under uncertainty.

The discrepancy between expected and observed sensor behavior highlighted the limitations of purely hardware-based approaches. The integration of the Kalman filter marked a conceptual turning point, shifting the focus from device functionality to system-level understanding. This experience reinforced the importance of combining engineering intuition with mathematical modeling when working with real-world systems.

Appendix

The full project repository, including:

- Arduino source code
- MATLAB implementation
- visual documentation (images and videos)
- intermediate development stages

is available at:

<https://github.com/sasokru/FinalProject25-26.git>

Eigenständigkeitserklärung

Hiermit versichere ich, dass ich die Arbeit – bei einer Gruppenarbeit den entsprechend gekennzeichneten Teil der Arbeit – selbstständig verfasst und keine anderen als die angegebenen Quellen und Hilfsmittel benutzt habe. Alle Stellen, die ich wortwörtlich oder sinngemäß aus fremden Werken entnommen habe, habe ich als solche kenntlich gemacht. Die Arbeit habe ich in gleicher oder ähnlicher Form noch keiner Prüfungsbehörde vorgelegt.

A handwritten signature in black ink, appearing to read 'S. Kruschinski', with a horizontal line underneath.

Sarah Sophie Kruschinski
Hamburg, 31.03.2026